

Phase 9 Cluster Diversity Visualization

Variant: baseline

1980 to 2025, every 5 years

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Single seed (seed=0), N=250, Independents=12%

Each dot = one agent. Position = (economic axis, cultural axis), both in [-1, 1].
Blue = Democrat, Red = Republican, Grey = Independent.
Saturation = identity strength. X marker = party centroid.

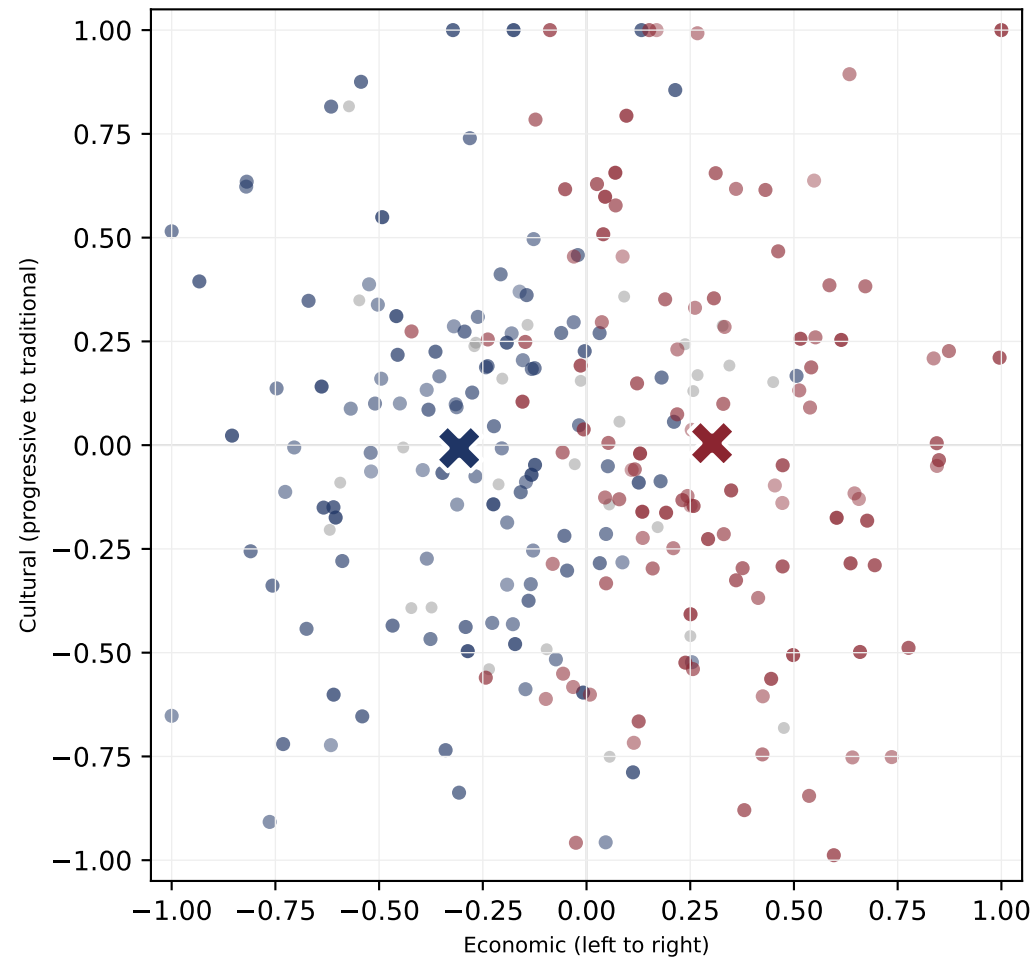
Key diagnostic metrics in each subplot title:
k* = effective number of clusters (silhouette-best k); larger = more cluster diversity
PR = participation ratio (covariance dimensionality)
bimY = bimodality on cultural axis (0..1; >0.55 = bimodal)
var, cons, aff = standard polsim metrics

Compare to the baseline historical-arc PDF (phase8f_visualization.pdf):
Baseline k* falls from 5.6 (1980) -> 2.0 (2000+) – visible cluster diversity collapses.
Phase 9 best variant aims to keep k* >= 3 across the arc.

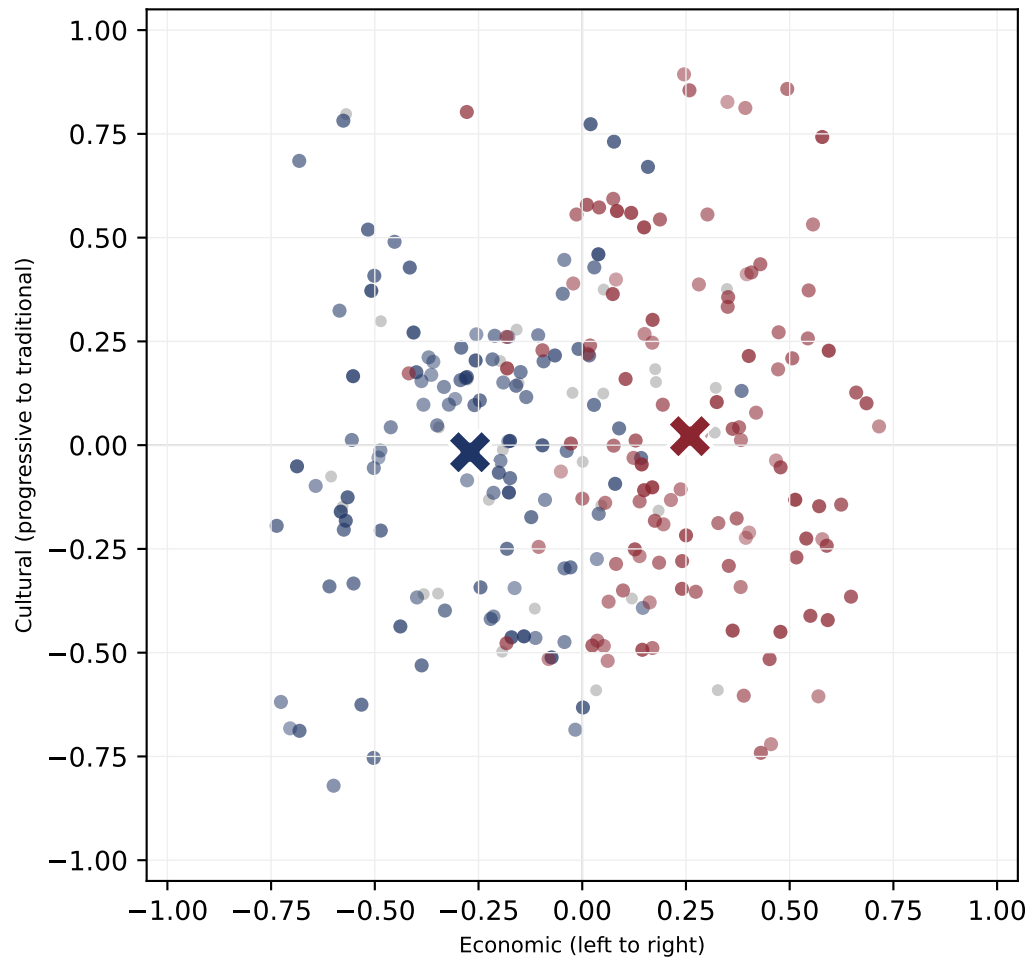
Faction set seeded at 1980 (from US_FACTIONS in scripts/phase9_cluster_diversity.py):

DSA_socialists (-0.65, -0.50)	progressive_liberals (-0.40, -0.55)
mainstream_dems (-0.30, -0.10)	blue_dog_dems (-0.15, +0.25)
evangelical_dems (-0.20, +0.40)	libertarian_reps (+0.55, -0.30)
classical_liberals (+0.25, -0.10)	mainstream_reps (+0.30, +0.15)
MAGA_populists (+0.45, +0.50)	sovereign_hard_right (+0.70, +0.65)
crunchy_centrists (-0.05, +0.10) – cross-pressured	

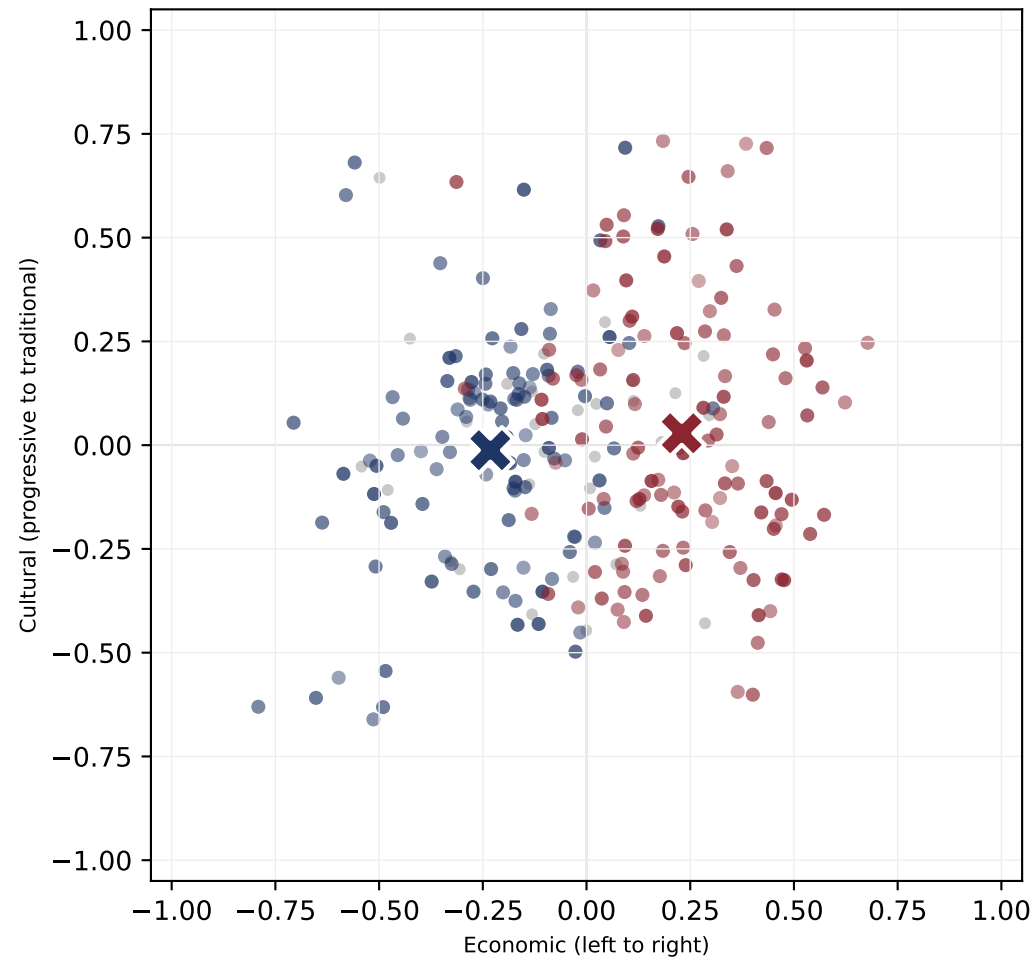
1980 $k^*=8.0$ PR=1.99 bimY=0.36
var=0.36 cons=0.37 aff=-0.25



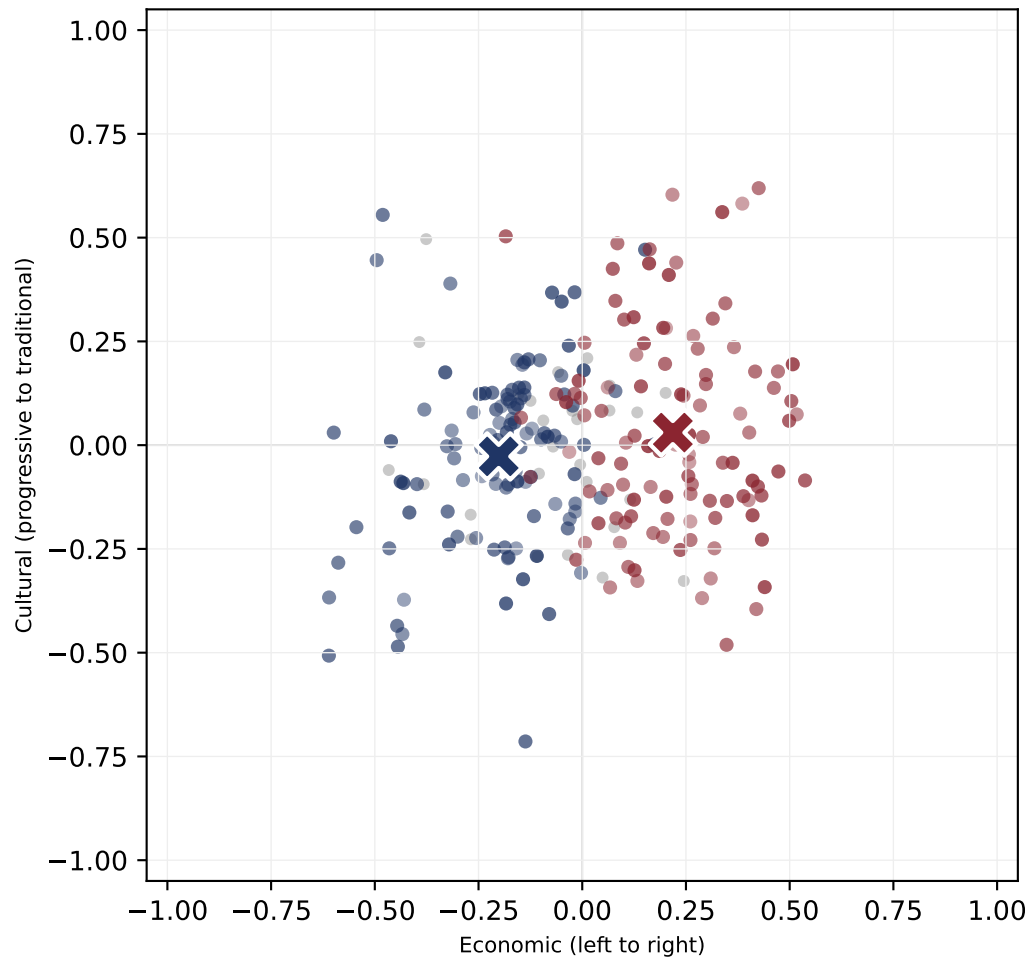
1985 $k^*=8.0$ PR=1.99 bimY=0.39
var=0.25 cons=0.40 aff=-0.38



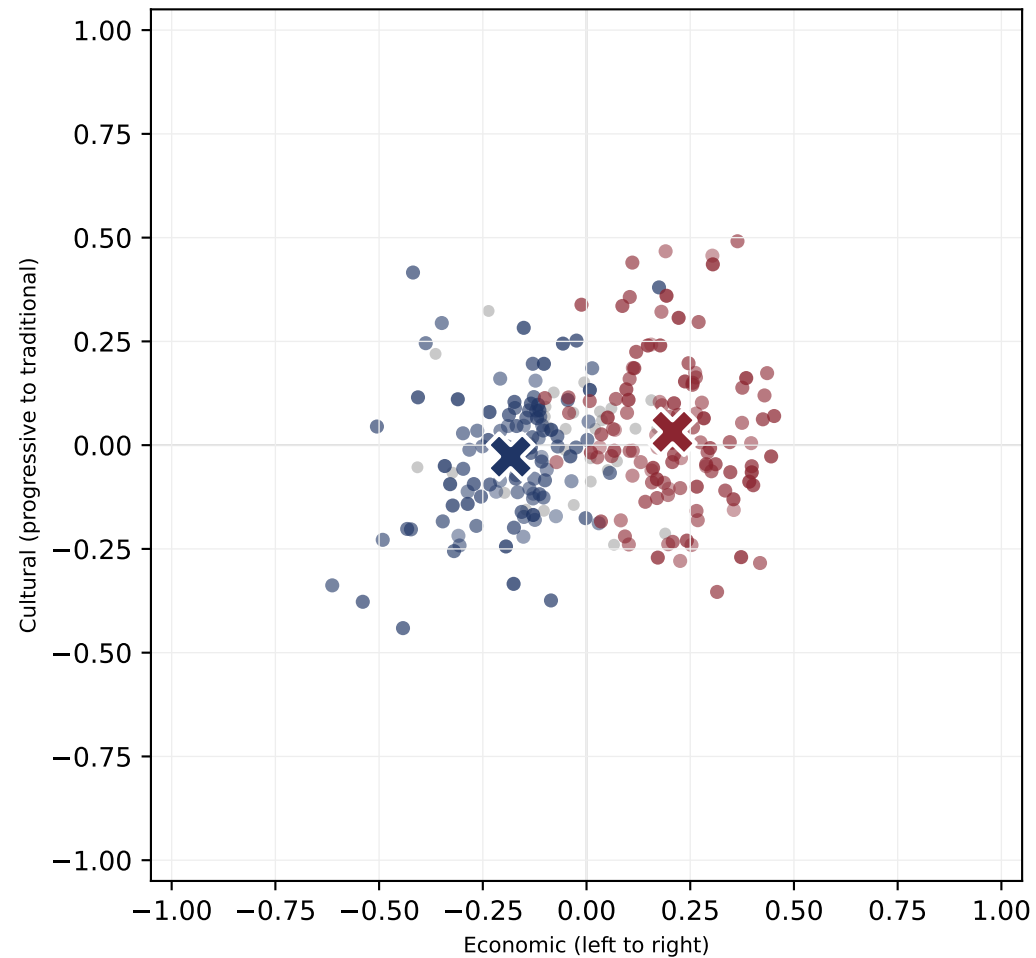
1990 $k^*=7.0$ PR=1.98 bimY=0.36
var=0.17 cons=0.41 aff=-0.49



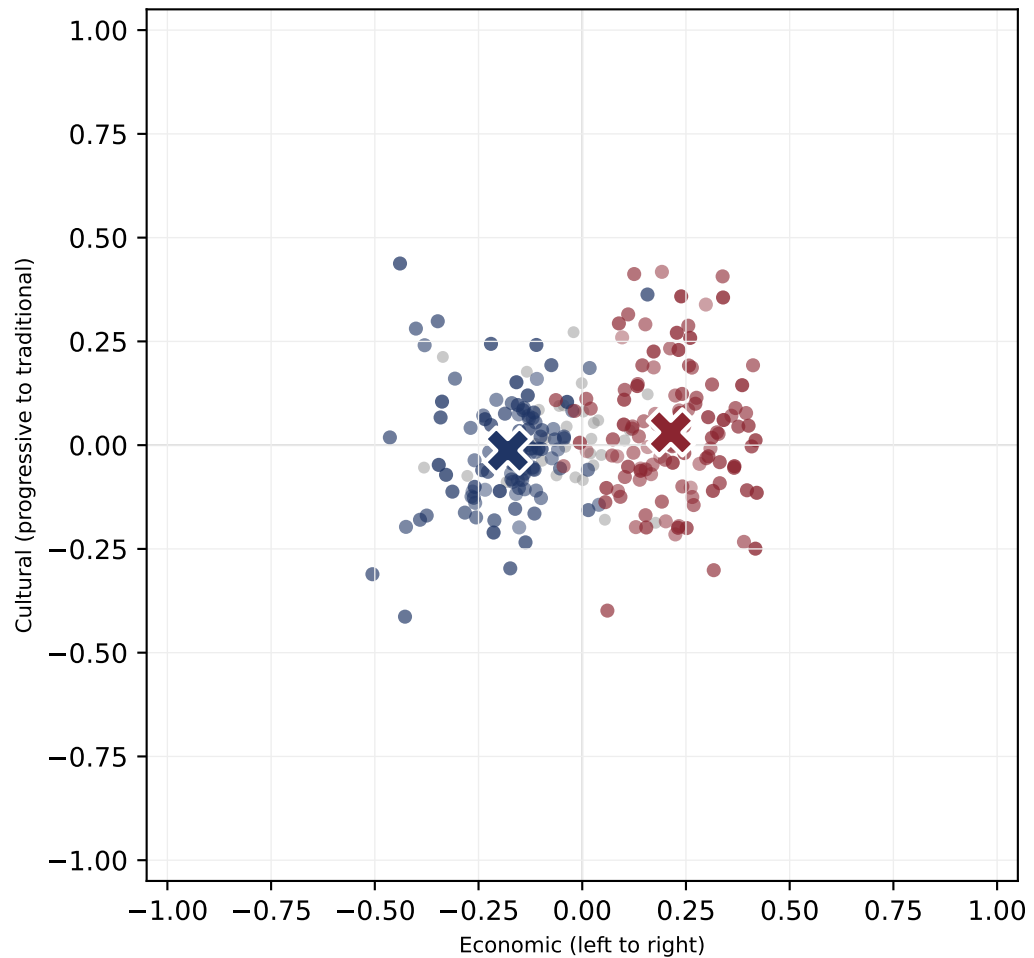
1995 $k^*=6.0$ PR=1.93 bimY=0.31
var=0.12 cons=0.46 aff=-0.57



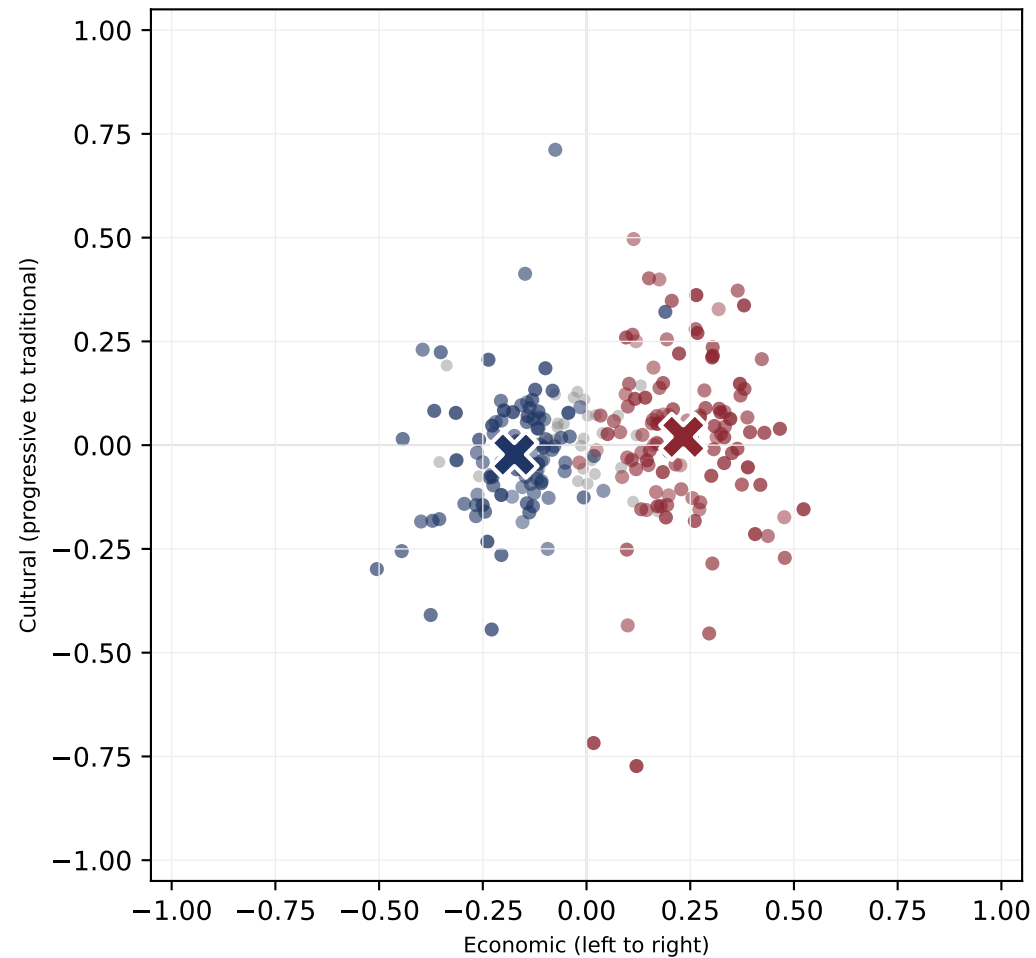
2000 $k^*=2.0$ PR=1.79 bimY=0.32
var=0.08 cons=0.51 aff=-0.63



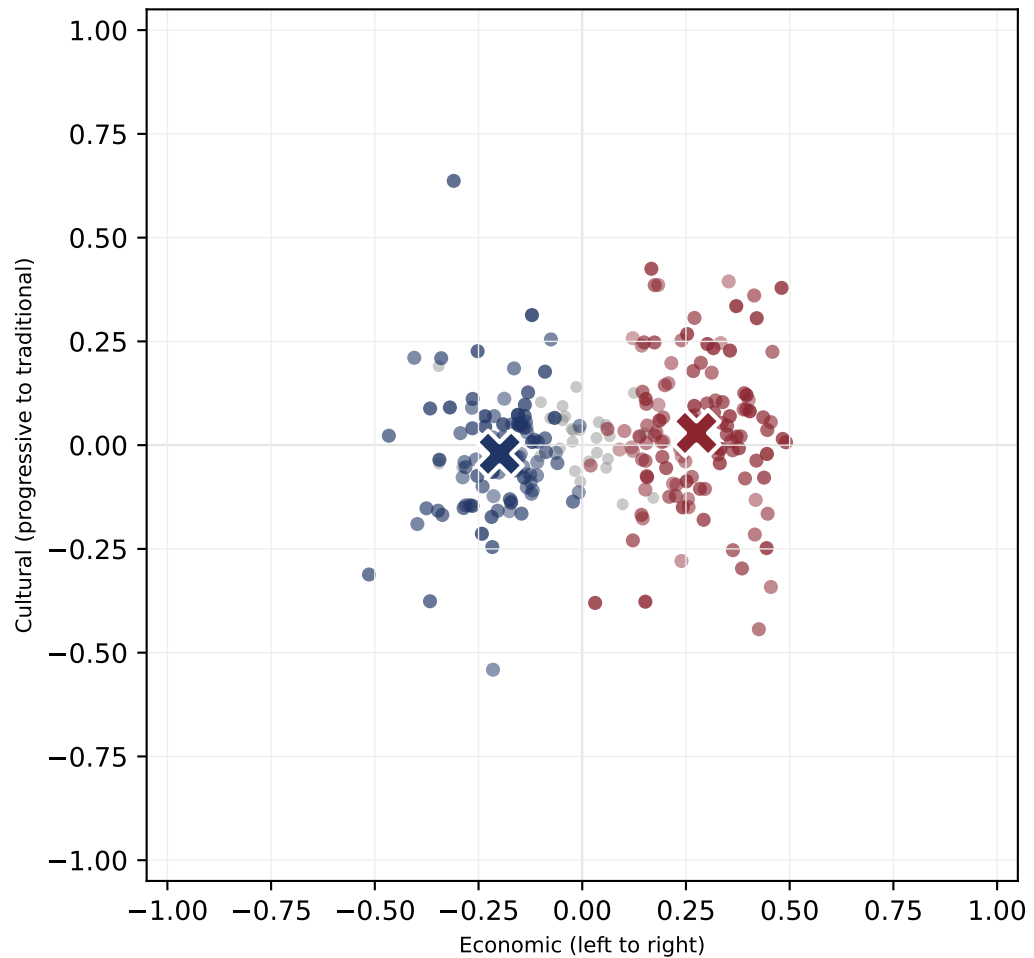
2005 $k^*=2.0$ PR=1.72 bimY=0.32
var=0.07 cons=0.51 aff=-0.68



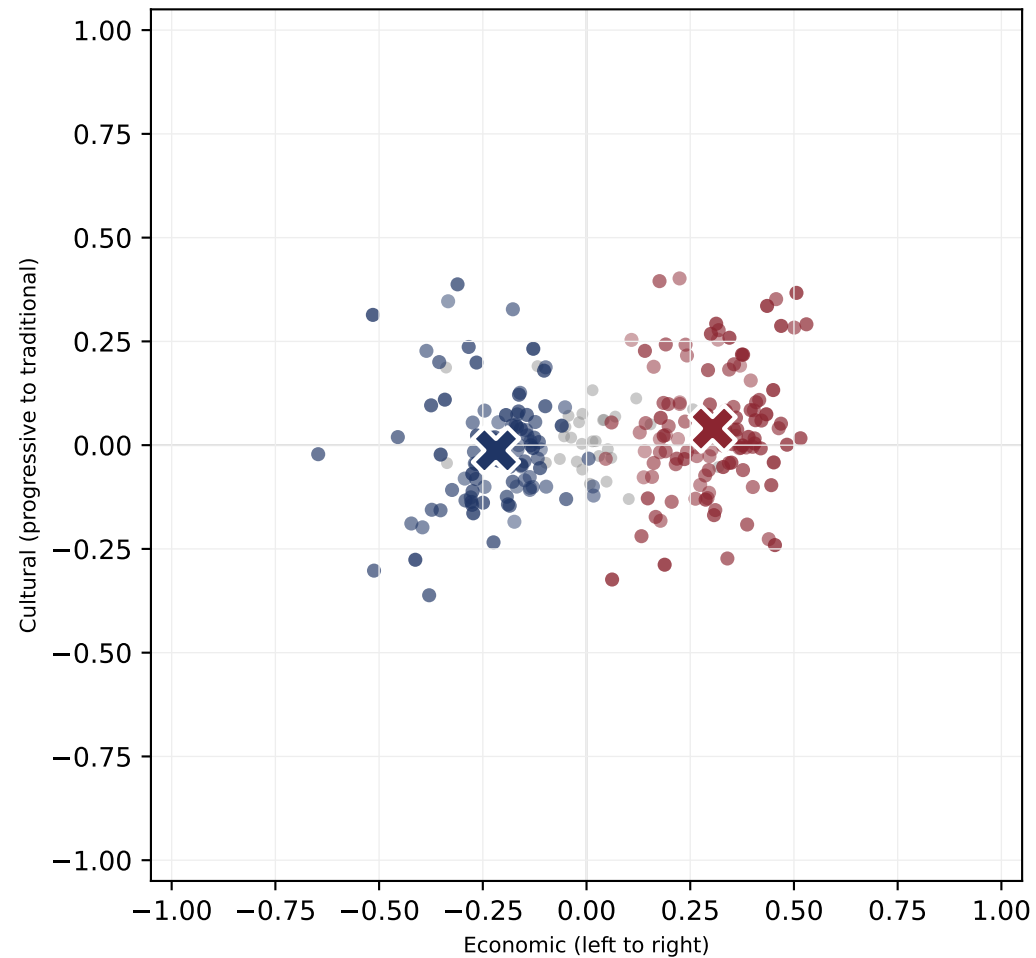
2010 $k^*=2.0$ PR=1.82 bimY=0.15
var=0.08 cons=0.50 aff=-0.69



2015 $k^*=2.0$ PR=1.64 bimY=0.21
var=0.08 cons=0.54 aff=-0.73



2020 $k^*=2.0$ PR=1.46 bimY=0.33
var=0.09 cons=0.55 aff=-0.77



2025 $k^*=2.0$ PR=1.41 bimY=0.31
var=0.10 cons=0.56 aff=-0.79

